

PP Pipe Products — Web Application

Project Document & Feature Guide

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Platform: Web Application (React + TypeScript)

Backend API: Node.js REST API (Port 8080)

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1. Executive Summary

PP Pipe Products is a comprehensive, cloud-based **Enterprise Resource Planning (ERP)** web application purpose-built for a pipe manufacturing company. It digitises and automates every aspect of the business — from manufacturing pipes on the factory floor all the way to invoicing customers and filing GST returns.

In plain language: Think of this as the company's complete digital brain. Every sale, every pipe produced, every bag of cement used, every payment received or made — it all gets recorded here in real time, giving management a single place to see everything that's happening in the business.

What the system covers:

- **Sales:** Retail counter sales (POS), B2B sales orders, invoices, quotations, payments
- **Purchases:** Vendor management, purchase orders, bills, payments to suppliers
- **Manufacturing:** End-to-end production tracking across 10 stages of pipe making
- **Inventory:** Real-time stock levels, adjustments, inter-outlet transfers
- **Daily Operations:** Cement consumption, vehicle usage, silo levels, loading records, testing lab, PDI
- **Finance:** GST reports, debtors, creditors, expense tracking
- **HR:** Staff management, role-based permissions, incentive calculations
- **Reporting:** 8 categories of reports with date filters, charts, and CSV exports

2. System Overview

Technology Stack

Layer	Technology	What it means
Frontend UI	React 18 + TypeScript	The screens users see and interact with
Styling	Tailwind CSS	Makes the app look clean and responsive
Charts	Recharts	Bar charts, pie charts, trend graphs
State Management	React Context + Hooks	Keeps data consistent across the app
API Communication	Axios	Sends/receives data to the backend server
Backend	Node.js REST API	The engine that processes all business logic
Database	PostgreSQL (via API)	Where all data is permanently stored

Application URL

- **Web App:** `http://localhost:3000` (development) / Company domain (production)
- **API Server:** `http://localhost:8080`

Multi-Outlet Support

The system supports **multiple factory outlets/branches**. Each user is assigned to an outlet, and data is filtered accordingly. An admin can view data across all outlets.

3. User Roles & Access Control

The system uses **Role-Based Access Control (RBAC)** — meaning each user only sees and can do what their role permits.

Roles Available

Role	Who it's for	Access Level
Super Admin	System owner / IT admin	Full access to everything, including settings
Admin	Factory manager / business owner	Full operational access
Manager	Department heads	Access to their module + reports
Accountant	Finance team	Sales, purchases, GST, reports
Staff	Counter operators, floor workers	Limited to their daily tasks

Permission Areas (Granular Control)

Admins can configure exactly what each role can do across 11 areas:

1. **Point of Sale** — Apply discounts, give refunds, void orders, manage shifts, accept payments
2. **Products & Catalogue** — Create, edit, delete products; manage categories

3. **Inventory** — View/adjust stock, approve transfers, receive stock
 4. **Customers** — Create, edit, view customer records
 5. **Orders** — Create, cancel, refund, view orders
 6. **Invoices** — Create, edit, send, delete invoices
 7. **Purchases** — Create POs, receive goods, pay vendors
 8. **Reports** — View financial reports
 9. **Expenses** — Create, approve expenses
 10. **Staff Management** — Add staff, reset passwords, deactivate accounts
 11. **Settings** — Modify system configuration
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4. Module 1 — Dashboard

URL: `/dashboard`

Who uses it: Managers, Admins

What it does

The Dashboard is the **home screen** after login. It shows a bird's-eye view of the business right now — no need to dig through multiple pages.

Key Features

Summary Cards (KPIs)

- **Today's Sales** — Total revenue collected today
- **Orders Count** — Number of orders processed
- **Production Summary** — Pipes manufactured across all stages
- **Intermediate Stock** — Work-in-progress inventory at each production stage

Date Filter

- 8 preset ranges: Today, Yesterday, This Week, Last Week, This Month, Last Month, This Quarter, This Year
- Custom date range picker for any specific period

Production Overview

- Shows pipe diameter breakdown (e.g., how many 600mm pipes, 800mm pipes)
- Stage-by-stage completion status with visual indicators
- Helps production managers know what's in progress at a glance

Charts & Visualisations

- Daily sales trend line
- Stage completion bar charts
- Intermediate stock table with quantities per stage

In plain language: The Dashboard is like a morning briefing — open it and you instantly know how yesterday went, what's happening today, and where production stands.

5. Module 2 — Point of Sale (POS)

URL: `/pos`

Who uses it: Counter staff, cashiers

What it does

The POS is the **retail counter billing system** — used when a customer walks in and buys pipes or materials directly. It works like a modern billing machine on a computer.

Key Features

Product & Customer Search

- Search products by name, SKU, or barcode
- Instantly find customers by name or phone number
- Add new customers on the spot if they're first-time buyers

Cart Management

- Add multiple products to the cart
- Change quantities with + / – buttons
- Remove individual items
- Hold an order (put it aside and come back to it later with a note)

Pricing & Discounts

- **Price Override:** Change the selling price for a specific item (with permission)
- **Item Discount:** Apply percentage or flat discount on individual items
- **Coupon Codes:** Apply promotional coupon codes for cart-level discounts
- **Credit Notes:** Apply previously issued credit notes to reduce the bill amount
- Automatic price list resolution (VIP customers get VIP pricing automatically)

Payment Processing

- Accept multiple payment modes: Cash, UPI, Card, Bank Transfer, Cheque, Split Payment
- Automatic change calculation for cash payments
- Payment confirmation screen with receipt option

Shift Management

- Cashiers open a **shift** at the start of the day (entering opening cash balance)
- Shifts track all transactions during that period
- At end of day, close the shift with a sales summary

Receipt Printing

- Printable receipt with company logo, GST details, items, total, and payment method
- Shareable digital receipt link

In plain language: The POS module is like a smart cash register — scan products, apply discounts, take payment, print bill. But it also keeps a complete record of every transaction automatically.

6. Module 3 — Sales

URL: /sales/*

Who uses it: Sales team, accountants, managers

What it does

The Sales module handles all **B2B (business-to-business) sales** — when pipes are sold to contractors, government projects, or other companies with formal documentation (sales orders, invoices, quotations).

Sub-modules

6.1 Orders

URL: /orders

Tracks all transactions — both POS retail and B2B orders.

- View order history with status badges: COMPLETED, PENDING, CANCELLED, HELD, REFUNDED
- Search by order number or customer name
- Summary stats: total orders, revenue, pending count
- Drill into any order for full line-item detail

6.2 Sales Orders (B2B)

URL: /sales-orders

A **Sales Order (SO)** is a formal document confirming what a customer has agreed to buy.

- Create new sales orders with multiple line items
- Link sales orders to production orders (so manufacturing starts automatically)
- Track status: DRAFT → CONFIRMED → DELIVERED → CANCELLED
- Generate invoice directly from the sales order
- Edit or cancel orders before fulfilment

6.3 Invoices

URL: /sales/invoices

Invoices are the **official billing documents** sent to customers requesting payment.

- Create invoices manually or auto-generate from sales orders or POS orders
- Status tracking: DRAFT, SENT, PAID, PARTIAL, OVERDUE, CANCELLED
- Record partial and full payments against an invoice
- Apply discounts and coupons on the invoice level
- GST-compliant tax calculation (CGST + SGST or IGST)
- **Send invoice by email** directly from the system
- Print invoice with company branding
- Shareable public invoice link (customer can view without logging in)

6.4 Quotations

URL: /sales/quotations

A Quotation is a **price estimate** sent to a customer before they confirm a purchase.

- Create formal quotations with product list and pricing
- Track status: DRAFT, SENT, ACCEPTED, EXPIRED, REJECTED
- Convert an accepted quotation into a sales order in one click
- Send quotation by email

6.5 Payments Received

URL: /sales/payments-received

Tracks all **money collected from customers**.

- View payment history by date, customer, and payment mode
- Status: PENDING, PARTIAL, COMPLETE

- Link payments to specific invoices

6.6 Sales Returns

URL: /sales/returns

When a customer returns goods — pipe is damaged, wrong size, etc.

- Record the return with reason
- Link to original invoice
- Automatic inventory adjustment on return acceptance

6.7 Credit Notes

URL: /sales/credit-notes

A Credit Note is issued when money is **owed back to the customer** (e.g., after a return or billing correction).

- Create credit notes linked to original invoices
- Apply credit notes against future invoices
- Cancel a credit note with reason

6.8 Delivery Challans

URL: /sales/delivery-challans

A Delivery Challan is a **transport document** that accompanies goods when delivered.

- Create and manage delivery challans
- Link to sales orders or invoices

In plain language: The Sales module is the complete paperwork trail from "customer wants to buy" to "money received" — quotation → order → invoice → payment → delivery.

7. Module 4 — Purchases

URL: /purchases

Who uses it: Purchase team, store managers, accountants

What it does

The Purchases module manages everything related to **buying materials, goods, and services** from vendors/suppliers.

Sub-modules

7.1 Vendors

Vendor is the database of all **suppliers** the company buys from.

- Add, edit, and deactivate vendor records
- Store contact details, GST number, payment terms
- Bulk import vendors from Excel/CSV
- Export vendor list

7.2 Purchase Orders

A Purchase Order (PO) is a **formal document sent to a vendor** to buy specific items at an agreed price.

- Create POs with multiple line items
- Track status: DRAFT → SENT → RECEIVED → BILLED → CANCELLED
- Receive stock against a PO (partial or full receipt)

7.3 Purchase Receives

Records the **physical receipt of goods** against a PO.

- Mark items as received with actual quantities
- Automatic inventory update on receipt
- Discrepancy flagging if received qty differs from ordered qty

7.4 Bills

A Bill is the **vendor's invoice to us** — what we owe the supplier.

- Create bills manually or convert from a PO
- Status: DRAFT, DUE, PARTIAL, PAID, OVERDUE
- Record payments against bills
- Track outstanding payables

7.5 Payments Made

Tracks all **money paid to vendors**.

- Payment history by vendor, date, and mode
- Link payments to specific bills

7.6 Vendor Credits

When a vendor owes us money (e.g., after returning defective goods).

- Create vendor credit notes
- Apply credits against future bills

7.7 Purchase Returns

When goods are **sent back to the vendor** due to defects or wrong supply.

- Record return with reason and quantity
- Linked to original purchase receive

7.8 Direct Purchase

For **quick, informal purchases** without going through the full PO workflow.

- Enter vendor, items, amounts directly
- Useful for urgent or one-time purchases

7.9 Bulk Purchase

For recording **large raw material intake** (e.g., cement, wire, aggregate in bulk).

- Record bulk quantities with conversion tracking
- Track conversion status: NOT_CONVERTED, PARTIALLY_CONVERTED, CONVERTED

In plain language: The Purchases module is the complete trail from "we need to buy something" to "we paid the supplier" — purchase order → goods receipt → vendor bill → payment.

8. Module 5 — Inventory

URL: /inventory

Who uses it: Store managers, production team, admins

What it does

Inventory keeps track of **what's in stock at all times** — raw materials, finished pipes, consumables.

Key Features

8.1 Stock View

- View current stock levels for every product
- Filter by: All Items, Raw Materials, PCCP Pipes
- Search by product name
- **Low stock alerts** — highlighted when stock falls below the reorder level
- Date range filtering to see stock movements over a period
- Export to CSV

8.2 Products Catalogue

URL: /products

The master list of all **products the company deals in**.

- Add, edit, delete products
- Each product has: Name, SKU, Barcode, Category, Unit of Measure, Purchase UoM, Reorder Level, Tax Group
- **Barcode generation and printing** — print shelf labels
- Bulk import from Excel template (with dry-run preview before import)
- Export product list to CSV/Excel
- Toggle product active / inactive

8.3 Categories

Organise products into **hierarchical categories** (parent → child).

- Create and manage product categories
- Toggle categories active/inactive

8.4 UoM Conversions

Unit of Measure (UoM) conversions — e.g., cement is purchased in tonnes but consumed in bags.

- Configure conversion factors per product
- Automatic conversion during transactions

8.5 Stock Adjustments

When physical stock count differs from system stock (e.g., after a manual audit).

- Record positive adjustments (stock found) or negative adjustments (stock missing)
- Every adjustment has an audit trail (who, when, reason)

8.6 Inter-Outlet Transfers

Move stock **from one outlet/branch to another**.

- Create a transfer request with items and quantities
- Receiving outlet approves and confirms receipt

- Automatic stock deduction/addition at each outlet

In plain language: Inventory is the company's digital stockroom. It always knows exactly what's available, flags when something is running low, and tracks every movement in and out.

9. Module 6 — Production

URL: `/production/*`

Who uses it: Production managers, floor supervisors, factory staff

What it does

The Production module tracks the **complete pipe manufacturing process** from raw materials to finished pipes across 10 manufacturing stages.

The 10 Production Stages

#	Stage	What happens
1	Fabrication	Steel cage/reinforcement is fabricated
2	Testing (Pre)	Cage tested for dimensions and quality
3	Moulding	Concrete is poured into the mould
4	Spinning	Pipe is spun to compact the concrete
5	Demoulding	Pipe is removed from the mould
6	Curing 1	First curing cycle (steam or water)
7	Curing 2	Second/extended curing
8	Winding	Pre-stress wire is wound (for PCCP pipes)
9	Coating	External protective coating applied
10	Final Testing	Finished pipe tested before dispatch

Sub-modules

9.1 Production Orders

URL: `/production/orders`

A Production Order (PO) is the **work order to manufacture a batch of pipes**.

- Create production orders specifying: pipe type, diameter, pressure class, quantity
- Link to customer sales orders (so you know who it's for)
- Status: DRAFT → PLANNED → IN_PROGRESS → COMPLETED → CANCELLED
- View production cost sheet per order
- Sortable by PO number, pipe name, planned quantity, status, dates

9.2 Production Entries

URL: `/production/entry`

A Production Entry records **actual work done** at a specific stage.

- Select the production order and stage
- Enter: quantity processed, machine used, shift name, bed type
- Record material consumption (cement, water, aggregate, wire) with actual quantities
- System verifies that the previous stage is complete before allowing the next entry
- Each entry creates an inventory deduction for materials consumed

9.3 Pipe Configurations

URL: /production/pipe-configs

The master configuration for each **type of pipe** the company manufactures.

- Define diameter (e.g., 300mm, 600mm, 1200mm) and pressure class (NP2, NP3, etc.)
- Set **material formula** — how much cement, aggregate, wire etc. goes into one pipe of each size
- Toggle configurations active/inactive

9.4 Machines

URL: /production/machines

Register all **production machinery** in the system.

- Add machines with type, name, and specifications
- Assign machines when recording production entries
- Toggle active/inactive

9.5 Shift Templates

URL: /production/shift-templates

Configure **named work shifts** (e.g., Morning 6AM–2PM, Afternoon 2PM–10PM, Night 10PM–6AM).

- Used when recording production entries
- Helps track which shift produced how many pipes

9.6 Overhead Configs

URL: /production/overhead-configs

Configure **indirect costs** to include in the production cost sheet.

- Electricity, water, machinery depreciation, labour overhead, etc.
- Each overhead is factored into the cost per pipe calculation

9.7 Production Reports

URL: /production/reports

- Stage-wise summary of how many pipes completed each stage
- Material consumption analysis (how much cement/wire used per period)
- Machine utilization metrics
- Cost summary per production order
- Filter by date range

In plain language: The Production module is like a factory logbook that never gets lost. Every pipe's journey from steel cage to finished product is tracked stage by stage, along with every bag of cement and every hour of machine time used.

10. Module 7 — Business Operations Hub

URL: `/business`

Who uses it: Factory supervisors, store keepers, lab staff, dispatch team

What it does

The Business Hub is a collection of **daily operational data entry modules** — everything that happens on the factory floor each day that doesn't fit neatly into sales or production but still needs to be tracked.

Sub-modules

10.1 Cement Bags

URL: `/business/cement-bags`

Track daily **cement consumption** on the factory floor.

- Enter date, quantity of bags used, notes
- View history with date filters
- Date picker limited to today or earlier (cannot log future dates)

10.2 Vehicles

URL: `/business/vehicles`

Track **company vehicle usage** — fuel filled, kilometres driven, purpose.

- Daily vehicle log entries
- Fleet overview

10.3 Diesel Maintenance

URL: `/business/diesel-maintenance`

Track **diesel purchases and maintenance costs** for vehicles and machinery.

- Record diesel quantity, amount, vehicle, date
- Separate tracking from general vehicle usage

10.4 Silo

URL: `/business/silo`

Track **aggregate silo levels** (sand, gravel stored in silos).

- Record current level per silo
- Date-wise history

10.5 Silo Extraction

URL: `/business/silo-extraction`

Record **material drawn from silos** for production.

- Quantity extracted, silo number, production order linked

10.6 Loading Records

URL: `/business/loading`

Record when pipes are **loaded onto trucks** for dispatch.

- Enter: date, vehicle number, customer site, pipe count, sizes
- Loading register for dispatch team

10.7 Loaded Pipes

URL: `/business/loaded-pipes`

View history of all **loaded pipe dispatches**.

- Filter by date range with preset pills
- Shows vehicle, customer, pipe counts

10.8 Transport Report

URL: `/business/transport-report`

Analyse **transport and logistics data** across three views:

- **By Vendor** — grouped by transport vendor with expandable details
- **By Customer** — grouped by delivery site/customer
- **All Trips** — flat list of every trip
- Search, filter, export CSV

10.9 Testing Lab

URL: `/business/testing-lab`

Record **quality test results** from the testing lab.

- Test date, pipe size, test type, result (Pass/Fail), remarks
- Linked to production batch

10.10 PDI (Pre-Delivery Inspection)

URL: `/business/pdi`

Pre-Delivery Inspection records before pipes leave the factory.

- Date, pipe reference, inspector, status (PASS/FAIL)
- 6 date range presets + custom date range picker
- Export records

10.11 Maintenance

URL: `/business/maintenance`

Log **equipment maintenance activities**.

- Machine, maintenance type, date, cost, notes

10.12 Store Room Material

URL: `/business/store-material`

Track **store room consumables** (nuts, bolts, tools, small materials).

- Issue and receipt entries
- Running balance per item

10.13 Labour

URL: `/business/labour`

Record **daily labour attendance and work**.

- Head count, work area, contractor name, date

10.14 Discard

URL: `/business/discard`

Log **rejected or damaged pipes** that cannot be sold.

- Quantity, reason, date, pipe configuration
- Linked to production order

10.15 Extra Fabrication

URL: `/business/extra-fab`

Record **additional fabrication work** outside of standard production orders.

10.16 Cutting

URL: `/business/cutting`

Track **pipe cutting operations** — when standard pipes are cut to custom lengths.

10.17 Conversion

URL: `/business/conversion`

Record **material conversion** — when bulk materials are broken into smaller units.

In plain language: The Business Hub is the factory's daily diary. Every bag of cement, every truck trip, every failed quality test — it all gets recorded here so management can track costs and spot problems.

11. Module 8 — Customers

URL: `/customers`

Who uses it: Sales staff, counter operators, admins

What it does

Maintains the **master database of all customers** — both retail and B2B.

Key Features

Customer List

- Search by name or phone number
- Customer **segment badges** with colour coding:
 - REGULAR (grey), SILVER (silver), GOLD (gold), VIP (purple), WHOLESALE (blue)
- Pagination for large lists
- Toggle customer active/inactive

Customer Record

Each customer stores:

- Name, phone, email, billing address, shipping address
- GSTIN (for B2B customers)
- Assigned price list / segment
- Purchase history

Bulk Operations

- **Bulk Import:** Upload an Excel/CSV file to add hundreds of customers at once
 - Dry-run preview: see what will be imported before committing
 - Download import template
- **CSV Export:** Export the full customer list

In plain language: The Customers module is the company's digital customer address book — but smarter, because it also remembers their segment, their purchases, and their outstanding dues.

12. Module 9 — HR & Staff

URL: `/staff` , `/incentives`

Who uses it: Admin, HR manager

12.1 Staff Management

URL: `/staff`

Manage all **system users** (employees who log into the system).

- Add new staff members with email, phone, role, outlet assignment
- Reset passwords
- Toggle active/inactive (deactivate without deleting)
- Assign **granular permissions** per staff member beyond their base role

Permissions Matrix

Each staff member can be fine-tuned across 58 individual permissions spanning 11 categories. Example:

- Can this cashier apply discounts? ✓ or ✕
- Can this manager delete invoices? ✓ or ✕
- Can this accountant view reports? ✓ or ✕

12.2 Incentives

URL: `/incentives`

Configure and calculate **staff performance incentives**.

- Set incentive rules (e.g., 1% of sales above ₹1,00,000)
- System automatically calculates monthly payouts per staff member
- Leaderboard showing top performers
- Monthly and yearly incentive tracking
- Recalculate button to refresh calculations

In plain language: The HR module controls who can log in, what they can see and do, and how much bonus they earn based on their performance.

13. Module 10 — Expenses

URL: /expenses

Who uses it: Accountants, managers, admins

What it does

Track and manage **all company expenditures** that aren't purchases of inventory — office expenses, utilities, travel, marketing, etc.

Key Features

Expense Records

- Record expenses with: amount, date, category, payment mode, description, reference number
- **Payment modes:** Cash, UPI, Bank Transfer, Card, Cheque, Other
- **Status workflow:** PENDING → APPROVED → PAID (or REJECTED)
- **ITC Eligibility** flag — mark if the expense qualifies for GST Input Tax Credit

Expense Categories

URL: /expenses/categories

- Create custom expense categories (e.g., Office Supplies, Travel, Utilities, Marketing)
- Set a **monthly budget** per category
- Customise colour and icon
- System alerts when spending approaches or exceeds the budget

Analytics

- Bar chart: expenses by category over time
- Pie chart: category-wise spend breakdown
- Monthly budget vs. actual comparison

Recurring Expenses

- Mark an expense as recurring (weekly, monthly)
- System auto-generates the next occurrence

Export

- Export expense records to CSV for accounting software

In plain language: The Expenses module is the company's expense register — digital, organised by category, with budget tracking and approval workflow built in.

14. Module 11 — Reports

URL: /reports/*

Who uses it: Managers, admins, accountants

What it does

The Reports module provides **data analytics and financial summaries** across 8 categories, all with date range filtering and export capabilities.

14.1 Reports Dashboard

URL: /reports

The starting point showing:

- Sales summary KPI cards (revenue, orders, customers)
- Daily sales trend chart
- Top 10 products by revenue
- Payment method breakdown
- Sales by category

14.2 Sales Report

URL: `/reports/sales`

- Revenue by day/week/month
- Product-wise sales breakdown
- Customer-wise sales summary
- 6 date presets + custom range
- Export to CSV

14.3 Purchase Report

URL: `/reports/purchases`

- Total purchases by vendor
- Outstanding purchase orders
- Goods received vs. billed analysis
- Payment due tracking

14.4 Inventory Report

URL: `/reports/inventory`

- Current stock levels across all outlets
- Stock movement (in/out) over a period
- Low stock alerts summary

14.5 GST Report

URL: `/reports/gst`

Compliance reporting for Goods & Services Tax filing.

Report	What it shows
GSTR-1	All outward (sales) supplies — what you sold and to whom
GSTR-3B	Summary of sales + ITC (Input Tax Credit) you can claim
HSN Summary	Sales and purchases grouped by HSN code (product classification)

- **Tally Export** — export in a format compatible with Tally accounting software
- **CSV Export** — for manual filing or other software

14.6 Payment Report

URL: `/reports/payments`

- All payments received by date, customer, payment mode
- UPI vs. cash vs. card breakdown
- Outstanding payment tracking

14.7 Debtors Report

URL: `/reports/debtors`

A Debtor is a **customer who owes money** to the company.

- List of all customers with outstanding invoices
- Amount overdue by each customer
- Aging analysis (how long the money has been outstanding)
- Date range filter + custom picker
- Export to CSV (amber Export button)

14.8 Creditors Report

URL: `/reports/creditors`

A Creditor is a **vendor the company owes money** to.

- List of all vendors with unpaid bills
- Amount due to each vendor
- Date range filter + custom picker
- Export to CSV

In plain language: Reports is the company's financial control centre — one place to see how sales are going, what the GST liability is, who owes you money, and who you owe money to.

15. Module 12 — Settings

URL: `/settings`

Who uses it: Admin, Super Admin only

What it does

System-wide configuration — only administrators can access this.

Settings Tabs

15.1 Factory (Outlet Configuration)

- Company name, code, phone, email, address
- **GST details:** GSTIN, state code, tax type (Regular/Composite)
- Multi-outlet: configure each branch separately

15.2 Roles

- Create and manage **user roles** (e.g., create a "Supervisor" role)
- Each role gets a custom name, description, and colour badge
- Assign permissions to roles

15.3 Permissions

The **master permission matrix** — visually assign which permissions each role has.

- Toggle individual permissions on/off per role
- Changes take effect immediately for all users with that role

15.4 Tax Groups

- Configure GST tax rates (5%, 12%, 18%, 28%)

- Create tax groups (e.g., "GST 18%" = CGST 9% + SGST 9%)
- Assign tax groups to products

15.5 Receipt Template

Customise the **printed receipt** from the POS:

- Company logo upload
- Header and footer message
- Show/hide fields (GST number, phone, address)
- Receipt width configuration

15.6 Document Templates

Customise **invoice and quotation layouts**:

- Upload company logo
- Set default template style
- Configure what fields appear on each document

15.7 Integrations

Connect the system to external services:

- **Email (SMTP):** Configure email server to send invoices and quotations directly
- Test email channel connectivity
- Future: SMS notifications, payment gateways

In plain language: Settings is the control panel for the whole system — only trusted admins use it to set up how the company's data appears on bills, who can do what, and how the system connects to email.

16. Module 13 — Activity Logs

URL: `/activity-logs`

Who uses it: Admin, Super Admin

What it does

The Activity Log is a **complete audit trail** of every action taken in the system. It answers the question: "Who did what, and when?"

Key Features

What gets logged

Every create, update, delete action across all 20+ modules:

- Invoice created, edited, deleted
- Customer added or modified
- Stock adjusted
- Payment recorded
- Staff password changed
- Settings modified
- Login and logout events

Filtering & Search

- Filter by **module** (e.g., show only Invoice-related actions)

- Filter by **action type** (Created / Updated / Deleted / Login / Logout)
- Filter by **user** (who performed the action)
- Filter by **date range**
- Search by keyword
- Paginated results

Visual Display

- Colour-coded action badges (green = Created, blue = Updated, red = Deleted)
- Exact timestamp of each action
- User name and role shown
- Changes recorded (what value changed from → to)

In plain language: Activity Logs is the security camera of the system — it records every action so if something goes wrong (a price was changed, a record was deleted), you can find out exactly who did it and when.

17. Technical Architecture

Frontend Architecture

```
src/
├── components/      # Reusable UI components (buttons, modals, pickers)
├── layouts/         # App shell with sidebar navigation (AppLayout.tsx)
├── pages/           # All page components (~96 pages across 16 directories)
│   ├── auth/       # Login
│   ├── business/   # 18 business operation pages
│   ├── customers/  # Customer management
│   ├── dashboard/  # Dashboard
│   ├── expenses/   # Expense tracking
│   ├── inventory/  # Stock management
│   ├── pos/        # Point of Sale
│   ├── production/ # Manufacturing
│   ├── purchases/  # Procurement
│   ├── reports/    # Analytics and reports
│   ├── sales/      # Sales and invoicing
│   ├── settings/   # System configuration
│   └── staff/      # HR and user management
├── providers/       # React Context: Auth, Cart state management
└── services/        # api.ts – all API calls centralised here
```

API Layer (`api.ts`)

All backend communication is centralised in a single service file with 40+ module groups:

- Axios instance with automatic auth token injection
- Token refresh handling on expiry
- Consistent error handling

State Management

- **AuthProvider** — user session, outlet selection, permissions
- **CartProvider** — POS cart state (items, quantities, discounts, customer)

- Local component state via `useState` / `useReducer` for forms and filters

Key UI Patterns Used Consistently

Pattern	Where used
Hero header with gradient	Every module page
Two-strip layout (tabs left, date filters right)	Reports, Inventory
6 date presets + CustomRangePicker	All date-filtered pages
Amber Export CSV button	All exportable reports
Collapsible expandable cards	Transport, Vendor summaries
Modal forms	Add/Edit entries across all modules
Toast notifications	All save/error actions

18. Key Benefits Summary

For the Business Owner

- **Real-time visibility** into sales, production, and financials from anywhere
- **GST compliance** — GSTR-1 and GSTR-3B ready to file every month
- **Debtor tracking** — instantly know which customers owe money and for how long
- **Cost control** — see exactly how much each pipe costs to produce

For the Factory Manager

- **Production tracking** — know exactly where every pipe is in the manufacturing process
- **Material consumption** — cement, wire, aggregate usage auto-tracked against production
- **Machine and shift records** — accountability for every production entry

For the Accounts Team

- **One system for everything** — no need to maintain separate Excel sheets
- **Automated reports** — one click to export GST reports, debtor lists, payment summaries
- **Audit trail** — every financial entry has a timestamp and username

For Counter Staff

- **Simple POS** — fast billing with barcode support
- **Shift management** — opening and closing cash tracked automatically
- **Customer lookup** — instantly find any customer by phone

For IT / Admin

- **Role-based security** — each employee only sees what they need
- **Activity logs** — full audit trail of all system changes
- **Bulk import** — onboard customers, vendors, products via Excel
- **Email integration** — send invoices and quotations directly from the system

Appendix — Page Count by Module

Module	Pages / Screens
Dashboard	1
POS	1 (+ 4 modals)
Sales	10
Purchases	9
Inventory	7
Production	12
Business Hub	18
Customers	4
HR & Staff	2
Expenses	2
Reports	9
Settings	2
Activity Logs	1
Auth / Profile	2
Total	~96 pages

*Document prepared for PP Pipe Products internal use.
For technical queries, contact the development team.*